

Tanush Singhal

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SUMMARY

Computer Science undergraduate and backend-leaning software engineer who delivers **end-to-end**. Two engineering internships (USA + Indian Railways) building **scalable services** on **AWS** and **PostgreSQL**, plus a 1st-place hackathon win. Strong in **Data Structures & Algorithms**, **System Design**, and ownership of production-quality, tested code.

EDUCATION

BITS Pilani – Pilani Campus <i>B.E. Computer Science Engineering (ID: 2023A7PS0586P)</i>	2023 – 2027
Indian Institute of Technology (IIT) Madras <i>B.S. Data Science and Applications (Online Degree)</i>	2023 – 2027

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, Bash
CS Fundamentals & Systems: Data Structures & Algorithms, System Design, Distributed Systems, Microservices, REST APIs, OOP, DBMS, Operating Systems, Concurrency, Scalability
Cloud, Backend & Tools: AWS (Lambda, API Gateway, S3, Bedrock), FastAPI, Node.js, Express.js, PostgreSQL, Docker, Git/GitHub, CI/CD, Operational Excellence, Linux

EXPERIENCE

Software Tree, LLC (USA) – Software Engineering Intern (Remote) <i>Python, LangChain, PostgreSQL, JDBC, JSON-RPC, Docker</i>	Oct 2025 – Present
– Owned end-to-end design of a 24/7 inventory-monitoring service (Python, event-driven agent over a JSON-RPC backend) that auto-detects low-stock items, computes reorder quantity, and dispatches alerts, replacing manual stock checks with hands-off automation. – Dove deep on database-access performance: designed and ran a 20-session controlled benchmark across 4 tasks on GPT-5.4 + Claude-4.6, instrumenting tokens, cost, and latency, and cut token (compute) cost by up to 58% for graph-shaped workloads. – Built a reusable Python JSON-RPC client with Server-Sent Events streaming and session handling, giving the team typed, schema-safe data access in place of raw SQL and simplifying integration. – Raised release quality by driving QA across 9 release iterations on live PostgreSQL / Oracle / MySQL / H2; root-caused and filed a reproducible defect with a workaround, surfaced 2 JDBC blockers, and integration-tested 4+ heterogeneous projects.	
Centre for Railway Information Systems (CRIS) – Software Development Intern <i>Node.js, Express, Sequelize ORM, SQLite, JWT, Docker Compose, React</i>	May 2025 – Jul 2025
– Designed and shipped a full-stack Parcel Management System end-to-end: 40+ REST APIs across 6 modules over a 5-table relational schema, JWT role-based access for 3 user roles, and Docker Compose deployment with durable volumes. – Engineered password-less OTP login with a 3-tier email failover (Gmail → Postmark → RapidAPI) that keeps authentication working when any single provider fails, plus a network-wide audit log for full traceability.	

PROJECTS

P2P Lending Voice AI Assistant (<i>1st of 50+ Teams</i>) <i>AWS Lambda, API Gateway, S3, Bedrock, WebSocket</i>	Jun 2025
– Led a 4-member team to design and deploy a serverless, event-driven assistant on AWS (API Gateway WebSocket → Lambda → Bedrock) with retrieval-grounded answers, token streaming for sub-second replies, and exponential-backoff retry plus a static fallback for reliability. – Delivered a working live demo in a 48-hour sprint, winning 1st place out of 50+ teams at the LenDenClub FinTech hackathon.	
AI Interview Integrity & Fraud Detection (<i>APOGEE 2026</i>) <i>Python, FastAPI, SQLite, scikit-learn</i>	Apr 2026
– Built a FastAPI backend with 30+ endpoints, 5 SQLite tables, and HMAC-verified webhooks, plus an explainable scoring engine (TF-IDF drift, readability, response-latency features) with a human-review safeguard that never auto-rejects a candidate.	

ACHIEVEMENTS & ACTIVITIES

1st Place / 50+ Teams – Matrix Protocol AI Hackathon (LenDenClub, FinTech): owned system design through AWS deployment to live demo in 48 hours.
Technical Author – Medium: published 2 long-form engineering articles (database-access performance, backend architecture) with companion YouTube and podcast walkthroughs, demonstrating learn-and-share depth.